

ANGELA R. LOPEZ SANCHEZ

+1 (202) 390-4849 | alopezsanchezv@gmail.com | [GitHub](#) | [Portfolio](#) | [Google Scholar](#)

EDUCATION

The University of Chicago

Master of Science in Computational Analysis and Public Policy

Chicago, IL

June 2026

Relevant Coursework: Advanced Machine Learning for Public Policy, Computer Science with Applications, Cloud Computing, Databases, Data Visualization

Award: Banco de la República (Central Bank of Colombia) National Merit Scholarship, one of only two awarded nationwide for graduate study in public policy

Universidad de los Andes

Master of Social Sciences in International Studies

Bogotá, Colombia

April 2018

Thesis: An Impact Evaluation of the Return of Colombians from Venezuela on Receiving Households' Well-being

Universidad Militar Nueva Granada

Bachelor of Arts in Economics

Bogotá, Colombia

October 2011

SKILLS

Programming Languages: Python, R, Stata, SQL, JavaScript, HTML. **Libraries & Frameworks:** D3.js, Altair/Vega, Dash, Next.js, MapLibre. **Cloud & Data Infrastructure:** AWS (S3, EC2, IAM), DuckDB, Supabase, Git/GitHub, reproducible CLI pipelines. **Geospatial & Remote Sensing:** Google Earth Engine, QGIS, ArcGIS, OSRM, FMM. **Methods:** Machine learning, causal inference & econometrics, geospatial analysis, microsimulation, survey data. **BI:** Power BI.

Languages: Spanish (Native), English (Fluent), French (Basic)

PROFESSIONAL EXPERIENCE

Inter-American Development Bank (IDB)

Senior Geospatial Consultant

Bogotá, Colombia (Remote)

May 2025 – Present

- After the M7.4 earthquake of August 2026 in Colombia, built and published a school emergency viewer covering 4,350 schools and 917,602 students in the affected zone, and a routing tool that planned inspections of 153 schools in two departments and served as the basis for costing and carrying out the inspection visits
- Architected and delivered the first regional school-accessibility dataset for Latin America and the Caribbean (22 countries, 530,000 schools), now used by IDB education teams to benchmark access gaps
- Built a reproducible Python CLI pipeline spanning geocoding, quality control, and dual travel-time modeling (FMM + OSRM network routing) to compute accessibility at continental scale

Organisation for Economic Co-operation and Development (OECD)

Data & Policy Consultant

Paris, France

June – September 2025

- Consolidated the first OECD social-capital dataset at subnational (TL2/TL3) resolution across OECD countries
- Produced statistical analysis and designed all data visualizations for Chapter 3, “Geographic Inequalities in Access to Opportunities,” of the flagship report *To Have and Have Not: How to Bridge the Gap in Opportunities*

The World Bank

Poverty Consultant

Washington, D.C.

August 2021 – August 2024

- Simulated poverty and inequality outcomes in Panama under alternative policy and macroeconomic scenarios using microsimulation models in Stata, informing the Bank’s Annual and Spring Meetings
- Evaluated the distributional impact of welfare programs (e.g., cash transfers) under alternative eligibility criteria to inform targeting of Panama’s beneficiary population
- Authored and contributed to 3 analytical products on poverty, inequality, gender gaps, and exclusion to support policy dialogue with the government, including a Systematic Country Diagnostic update and a Poverty Assessment in which I led the labor-market background chapter

Inter-American Development Bank (IDB)

Data & Policy Consultant

Washington, D.C.

June – November 2022

- Consolidated the first regional dataset on learning-assessment characteristics for Latin America and the Caribbean
- Drafted and delivered the policy report *The State of Education in Latin America and the Caribbean: Learning Assessments*, supporting evidence-based education-policy dialogue

Inter-American Development Bank (IDB)

Economic Analyst, Education Division

Washington, D.C.

July 2018 – August 2021

- Produced education and social statistics for LAC countries from household surveys and administrative data, delivering analytical

products that supported policy dialogue with regional governments

- Provided analytical and statistical support to launch the Bank's new Higher Education agenda for Latin America
- Built data tools (a dashboard and automated reports) and advised the Bank's Social Sector divisions on recurring data analysis

Departamento Administrativo Nacional de Estadística (DANE – Colombia National Statistics Office)

Bogotá, Colombia

Coordinator, SDG Unit

February – June 2018

- Led a team of 4 across DANE departments to define and design the methodology for 56 national indicators tracking the Sustainable Development Goals
- Coordinated with external agencies to meet Post-2015 Agenda reporting requirements, and represented Colombia's statistics office in 3 international meetings setting SDG indicator methodologies

Departamento Administrativo Nacional de Estadística (DANE)

Bogotá, Colombia

Economic Analyst, Labor Market Unit

February 2015 – January 2018

- Produced monthly analyses of Colombia's labor market and socio-demographic indicators for national and international dissemination
- Built and processed monthly household-survey datasets underpinning official labor-market and socioeconomic statistics, and trained survey enumerators quarterly to ensure data quality

Universidad Militar Nueva Granada

Bogotá, Colombia

Research Assistant

February 2013 – December 2014

- Co-authored 7 national and 3 international academic presentations, 3 published articles, and 8 virtual learning objects
- Led weekly sessions of the Economic Studies Research Group and advised undergraduates on research-initiation projects

SELECTED PROJECTS

- **EduAccess LAC, School Accessibility Platform:** Shipped a full-stack geo-platform (Next.js, Supabase Postgres, Railway cron worker) that turns the IDB accessibility indicators for five LAC countries into a tool a non-technical ministry director can use; built a two-tier LLM cascade (Llama 3.1/3.3) that converts plain-English questions into prompt-injection-guarded, validated SQL and highlights results on a MapLibre map ([Live](#))
- **Poverty Classification with Multi-Modal Data:** Trained a machine-learning model to sort households into poverty categories from satellite imagery plus a few proxy-means-test questions, a lower-cost way to target social programs where survey data is missing (Python, ML) ([Repo](#))
- **Panama Life-Cycle Activity Explorer:** Interactive visualization of how participation in education, work, and retirement shifts across age, gender, and region in Panama, surfacing where youth leave school or women exit the labor force (Python, D3.js) ([Live](#))

For a complete list of projects, see my [portfolio](#).

SELECTED PUBLICATIONS

Peer-Reviewed Journal Articles

- López, A.R., Okoye, K., Haruna, H., et al. (2022). Impact of Digital Technologies upon Teaching and Learning in Higher Education in Latin America: an Outlook on the Reach, Barriers, and Bottlenecks. *Education and Information Technologies*
- López, A.R., Virgüez, A.F., Silva, A.C., & Sarmiento, J.A. (2017). Desigualdad de oportunidades en el sistema de educación pública en Bogotá, Colombia. *Lecturas de Economía*, (87), 165–190
- López, A.R., Virgüez, A.F., Sarmiento Espinel, J.A., & Silva Arias, A.C. (2015). El efecto de la gerencia privada de escuelas públicas en el desempeño estudiantil en la educación media en Colombia. *Ecos de Economía*, 19(41), 108–136

Selected Policy Briefs

- López, A. (2024). *Panama Poverty Assessment: Technical Note on Labor Market*. The World Bank, Washington, D.C.
- Arias, E., López, A., Dueñas, X., & Giambruno, C. (2024). *The State of Education in Latin America and the Caribbean: Learning Assessments*. Inter-American Development Bank, Washington, D.C.
- Arias, E., López, A., Elacqua, G., et al. (2021). *Educación superior y COVID-19 en América Latina y el Caribe: Financiamiento para los estudiantes*. Inter-American Development Bank, Washington, D.C.

For a complete list of publications, see my [Google Scholar profile](#).